

# MASTER-US research terminal

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## F1 · MONITOR

### Does the market-guided gate add anything over the ungated architecture?

hypothesis · MSTR.US vs MSTR.NG · 5 seeds per arm per budget

0.949

→

0.464

ratio · short 12/4

ratio · full 100/10

NOT DISTINGUISHABLE

verdict

A ratio below 1.0 means the gap sits inside pooled seed dispersion. Phase 3's **0.949** was a 5% near-miss — the shape of a real effect masked by too little power. Training both arms to the spec's full schedule **doubled the gap** and **more than doubled the dispersion**, moving it further inside the noise. A real effect behaves the opposite way.

### Variants

click a row → QUOTE · headers sort

| TICKER  | VARIANT                  | RANKIC          | ICIR  | L/S GROSS | L/S NET     | Δ NET VS MSTR.US |
|---------|--------------------------|-----------------|-------|-----------|-------------|------------------|
| MSTR.US | Full MASTER              | +0.0212 ±0.0009 | 0.122 | 0.77      | -0.68 ±0.12 | reference        |
| MSTR.NG | No gating                | +0.0201 ±0.0006 | 0.112 | 0.71      | -0.74 ±0.10 | -0.058           |
| MSTR.NX | No inter-stock attention | +0.0211 ±0.0014 | 0.118 | 0.76      | -0.62 ±0.05 | 0.064            |
| MSTR.TA | Time-aligned             | +0.0205 ±0.0007 | 0.117 | 0.82      | -0.58 ±0.11 | 0.099            |
| MSTR.SH | Market vector shuffled   | +0.0207 ±0.0002 | 0.109 | 0.60      | -0.64 ±0.14 | 0.038            |
| LGBM.BL | LightGBM (tuned)         | +0.0207 ±0.0007 | 0.142 | 0.83      | -0.89 ±0.16 | -0.214           |
| LGBM.SP | LightGBM (shipped cfg)   | +0.0185 ±0.0005 | 0.147 | 0.95      | -1.17 ±0.20 | -0.494           |
| LSTM.BL | Per-stock LSTM           | +0.0139 ±0.0034 | 0.074 | 0.38      | -0.28 ±0.10 | 0.396            |
| RIDG.BL | Cross-sectional ridge    | +0.0162 ±0.0000 | 0.110 | 0.99      | -0.89 ±0.00 | -0.210           |

### Cost cascade

MSTR.US

### Build

from cached PhaseResults

gross L/S



0.77



0

DATA

298 tests · leakage gates pass

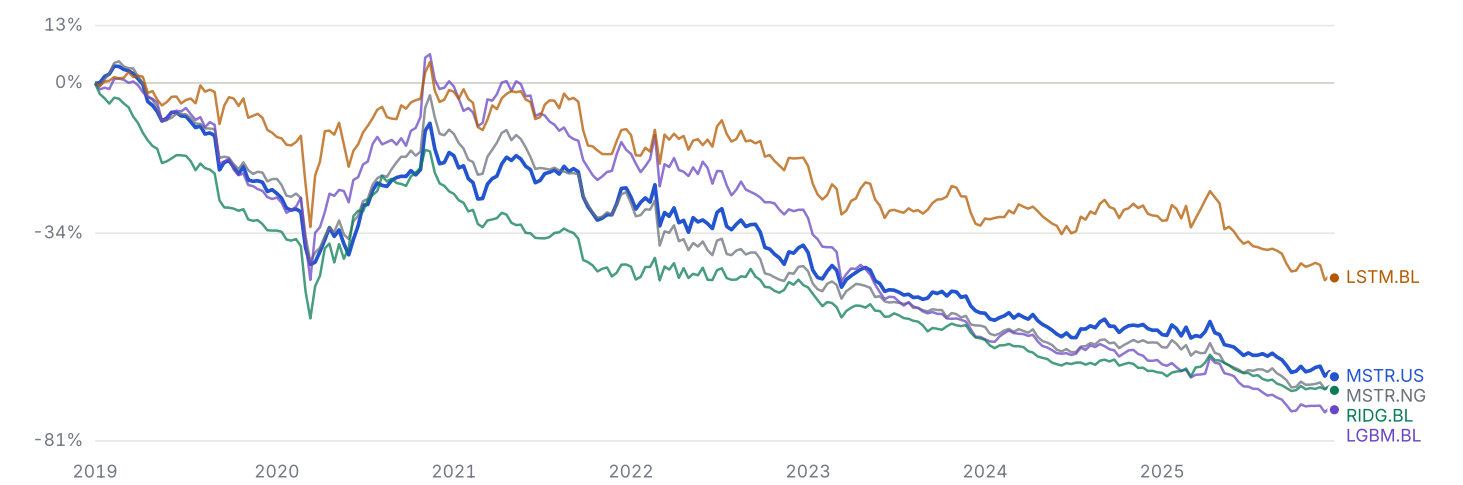
net @10 bps  -0.68

At the 10 bps baseline the dollar-neutral book is **net-negative** for every variant measured. Gross alpha exists; daily rebalancing at 116% turnover consumes all of it.

|   |   |          |                                       |
|---|---|----------|---------------------------------------|
| ✓ | 1 | ENGINE   | momentum Sharpe -0.13 · 2009...       |
| ✓ | 2 | BASELINE | LGBM RankIC +0.0207                   |
| ✓ | 3 | MASTER   | RankIC +0.0212 ±0.0009                |
| ✓ | 4 | ABLATION | gate null settled · ratio 0.949 →...  |
| ✓ | 5 | FACTORS  | market 14.2%/yr · R <sup>2</sup> 0.26 |
| ✓ | 6 | RISK     | bias gate 94.6% in [0.9, 1.1]         |
| ✓ | 7 | JOIN     | net -0.68 → neutralized -0.91         |
| ○ | 8 | REPORTS  | not started                           |

Equity — cumulative net return

test split 2019–2025 · seed-averaged



net of costs at the 10 bps baseline, dollar-neutral decile book, seed-averaged

MSTR.US

Full MASTER · isolates: — reference

-0.68

net L/S Sharpe

0.77

gross

±0.122

seed dispersion

116%

turnover

reference

vs MSTR.US

Metrics

gross / net pair · every figure over 5 seeds

| METRIC        | GROSS          | NET         | N |
|---------------|----------------|-------------|---|
| RankIC        | 0.0212 ±0.0009 | —           | 5 |
| ICIR          | 0.122 ±0.005   | —           | 5 |
| L/S Sharpe    | 0.77 ±0.13     | -0.68 ±0.12 | 5 |
| Breakeven bps | 10.6 ±1.6      | —           | 5 |

Seed dispersion

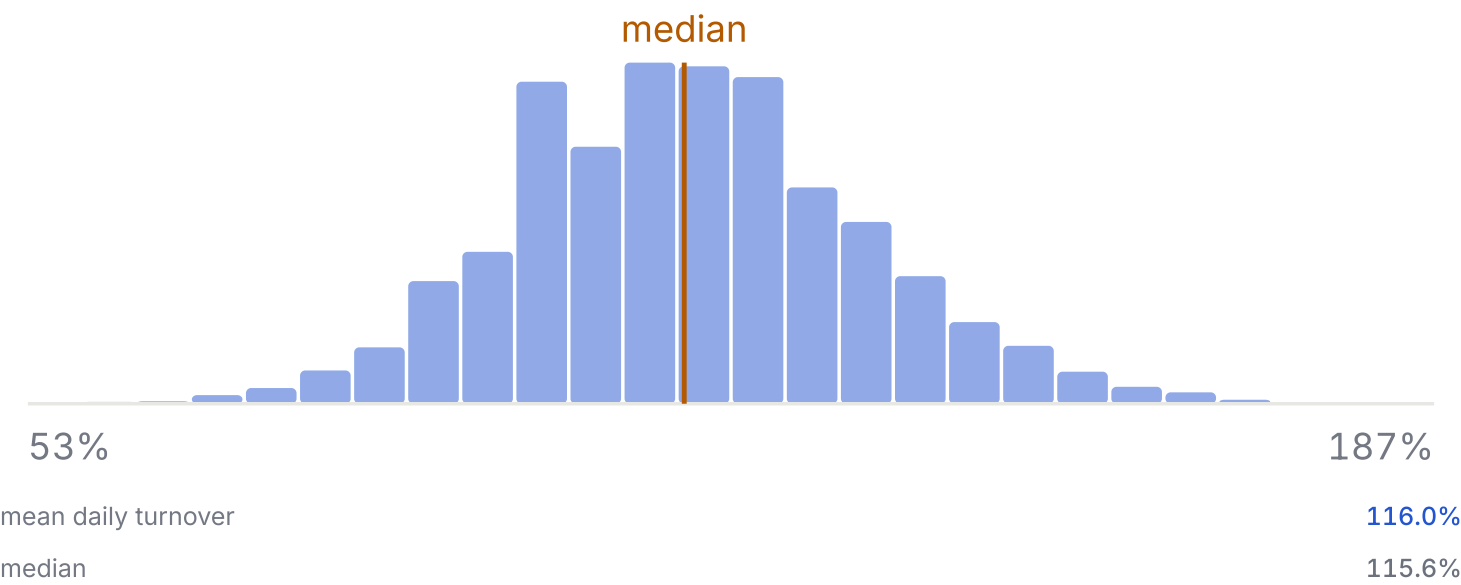
seeds 0, 1, 2, 3, 4



Each dot is one initialization; the band is ±1 SD around the mean. Tight clusters and wide scatters both average to a point estimate — the strip is what keeps that difference visible, and it is the difference every distinguishability call turns on.

Turnover distribution

daily, pooled across seeds



Stress — \$8.5

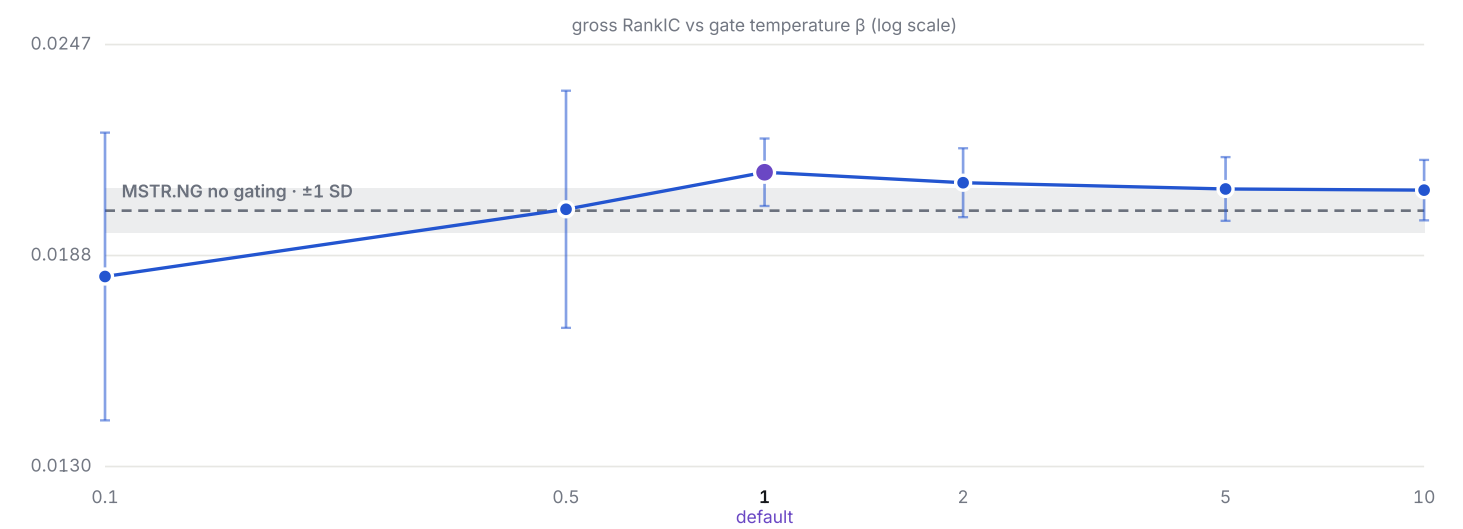
seed-averaged scores

| SLICE                | RANKIC             | TEST DATES |
|----------------------|--------------------|------------|
| 2018Q4 selloff       | outside test split | 0          |
| COVID crash          | +0.0028            | 62         |
| 2022 bear            | +0.0264            | 251        |
| 2023-2025 recovery   | +0.0158            | 751        |
| cap tier · large     | +0.0211            | —          |
| cap tier · mid       | +0.0201            | —          |
| cap tier · small     | +0.0257            | —          |
| sector-neutralized   | +0.0177            | —          |
| signal decay · h=1d  | +0.0200            | —          |
| signal decay · h=3d  | +0.0160            | —          |
| signal decay · h=5d  | +0.0118            | —          |
| signal decay · h=10d | +0.0083            | —          |

| SLICE                | RANKIC  | TEST DATES |
|----------------------|---------|------------|
| signal decay · h=21d | +0.0058 | —          |

β sweep — gate temperature against the no-gating reference

§8.2 · 6 temperatures × 5 seeds · the experiment that settles the question



No temperature separates from no-gating, and the curve is flat above  $\beta=0.5$ . Two coherence checks make that flatness read as measurement rather than noise: the **high-β limit converges to the reference** exactly as theory requires ( $\beta \rightarrow \infty$  approaches no gating), and **hard selection actively hurts** —  $\beta=0.1$  falls below the reference with several times the seed dispersion. The gate's only clearly detectable effect is harm when forced toward hard selection.

Architectural ablations

each cell differs from MSTR.US by exactly one thing

| TICKER  | ISOLATES                                     | ΔRANKIC | GROSS               | ΔNET L/S | NET                 |
|---------|----------------------------------------------|---------|---------------------|----------|---------------------|
| MSTR.NG | the market gate                              | -0.0011 | not distinguishable | -0.058   | not distinguishable |
| MSTR.NX | cross-sectional modelling                    | -0.0000 | not distinguishable | +0.064   | not distinguishable |
| MSTR.TA | the cross-time claim                         | -0.0007 | not distinguishable | +0.099   | not distinguishable |
| MSTR.SH | whether the gate reads real market structure | -0.0004 | not distinguishable | +0.038   | not distinguishable |

β sweep · figures

vs the no-gating reference

| β     | RANKIC          | VS REF  | NET L/S | BE   | VERDICT             |
|-------|-----------------|---------|---------|------|---------------------|
| 0.1   | +0.0183 ±0.0040 | -0.0018 | -0.62   | 8.2  | not distinguishable |
| 0.5   | +0.0201 ±0.0033 | +0.0000 | -0.63   | 10.0 | not distinguishable |
| 1 def | +0.0212 ±0.0009 | +0.0011 | -0.68   | 10.6 | not distinguishable |
| 2     | +0.0209 ±0.0010 | +0.0008 | -0.66   | 10.5 | not distinguishable |
| 5     | +0.0207 ±0.0009 | +0.0006 | -0.64   | 10.7 | not distinguishable |
| 10    | +0.0207 ±0.0008 | +0.0006 | -0.65   | 10.3 | not distinguishable |
| ref   | +0.0201 ±0.0006 | —       | -0.74   | —    | MSTR.NG no gating   |

Gate null · both training budgets

ratio < 1.0 ⇒ gap inside pooled seed dispersion

| BUDGET      | MEASURE        | GATED           | UNGATED         | GAP     | POOLED SD | RATIO | VEI |
|-------------|----------------|-----------------|-----------------|---------|-----------|-------|-----|
| short 12/4  | gross RankIC   | 0.0212 ±0.0009  | 0.0201 ±0.0006  | +0.0011 | 0.0011    | 0.949 | no  |
|             | net L/S Sharpe | -0.6806 ±0.1221 | -0.7383 ±0.0967 | +0.0577 | 0.1557    | 0.371 | no  |
| full 100/10 | gross RankIC   | 0.0224 ±0.0022  | 0.0210 ±0.0020  | +0.0014 | 0.0030    | 0.464 | no  |
|             | net L/S Sharpe | -0.4556 ±0.2611 | -0.5590 ±0.2731 | +0.1034 | 0.3778    | 0.274 | no  |

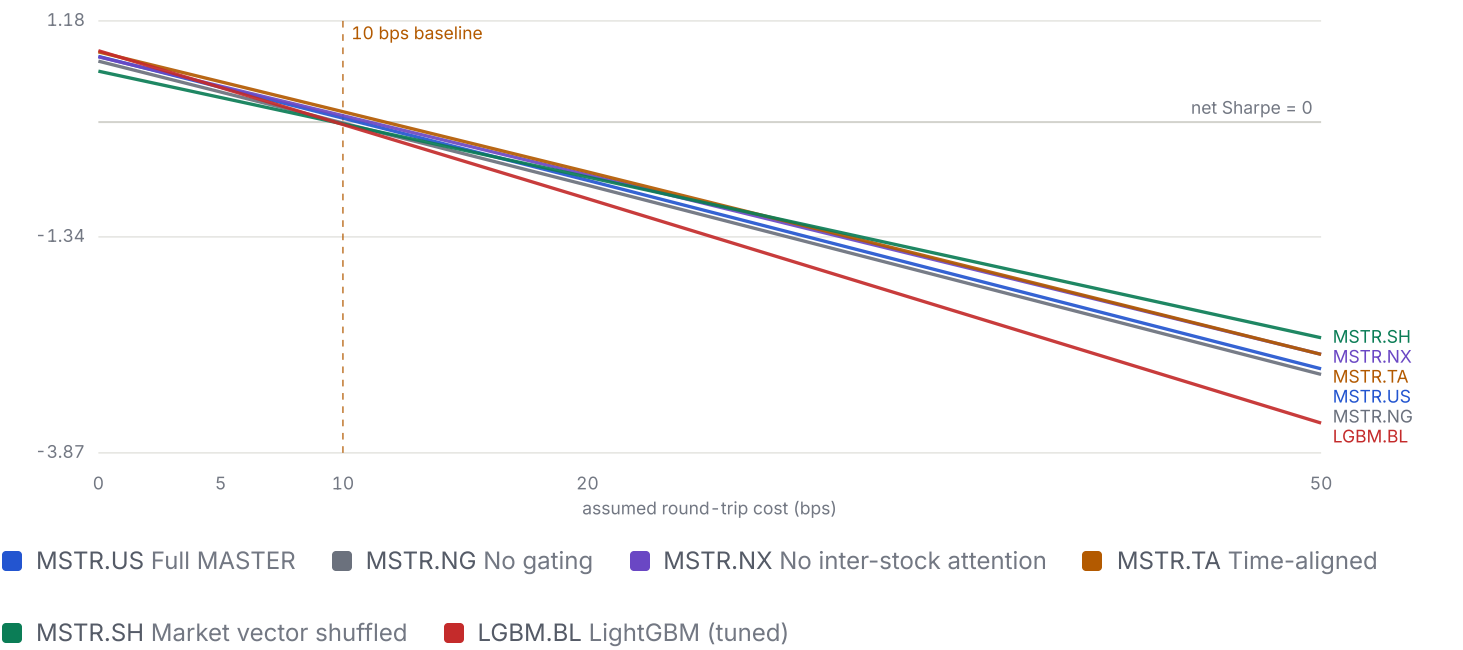
Reading the grid

None of MASTER's three structural mechanisms is distinguishable from its ablation. Removing inter-stock attention changes RankIC by -0.0000. And MSTR.SH — the gate fed a date-permuted market vector, its entire input destroyed while every marginal distribution and cross-feature correlation is preserved — costs -0.0004. The gate is not reading market structure.

What would legitimately re-open this: an architectural change to the gating mechanism itself, or a materially different experimental setup — different universe, label horizon, or feature bank. Not more seeds and not more epochs; both were tried, and the ratio moved the wrong way.

Net Sharpe vs assumed cost

dollar-neutral decile book · the analysis the original paper omitted



Breakeven

bps at which net alpha reaches zero

| TICKER  | VARIANT                  | BREAKEVEN | TURNOVER | @0   | @5   | @10   | @20   | @50   |
|---------|--------------------------|-----------|----------|------|------|-------|-------|-------|
| MSTR.TA | Time-aligned             | 11.8      | 117%     | 0.82 | 0.47 | 0.12  | -0.58 | -2.72 |
| MSTR.NX | No inter-stock attention | 11.0      | 113%     | 0.76 | 0.42 | 0.08  | -0.62 | -2.71 |
| MSTR.US | Full MASTER              | 10.6      | 116%     | 0.77 | 0.41 | 0.05  | -0.68 | -2.88 |
| RIDG.BL | Cross-sectional ridge    | 10.5      | 129%     | 0.99 | 0.52 | 0.05  | -0.89 | -3.74 |
| LSTM.BL | Per-stock LSTM           | 9.9       | 55%      | 0.38 | 0.21 | 0.05  | -0.28 | -1.28 |
| MSTR.NG | No gating                | 9.8       | 117%     | 0.71 | 0.35 | -0.01 | -0.74 | -2.95 |
| LGBM.BL | LightGBM (tuned)         | 9.7       | 134%     | 0.83 | 0.40 | -0.03 | -0.89 | -3.52 |
| MSTR.SH | Market vector shuffled   | 9.5       | 107%     | 0.60 | 0.29 | -0.02 | -0.64 | -2.52 |
| LGBM.SP | LightGBM (shipped cfg)   | 9.0       | 140%     | 0.95 | 0.42 | -0.11 | -1.17 | -4.39 |

## Why this panel exists

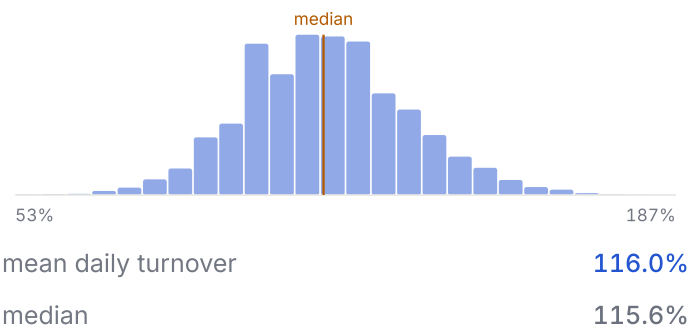
Every variant breaks even between **9.0 and 11.8 bps** against a **10 bps** baseline assumption. The whole family straddles the cost line: at 10 bps the net Sharpes cluster around zero, and by 20 bps everything is decisively underwater. A gross-only table would have shown a set of plausible alpha models.

Cost model: Corwin-Schultz, overnight-gap adjusted, plus sqrt market impact. Turnover is a permanent column throughout this project, never an optional one — the two orderings genuinely differ, and a model can rank last on gross RankIC while ranking first on net.

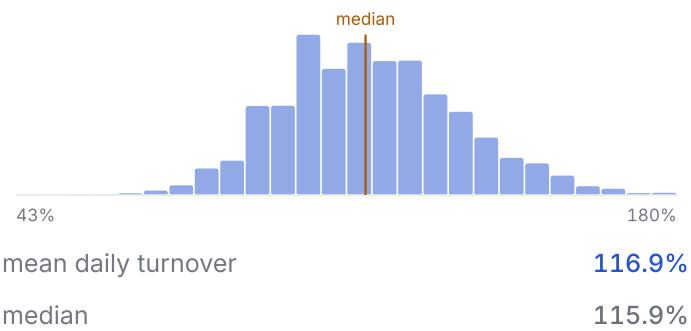
## Turnover distributions

daily one-way turnover, pooled across seeds

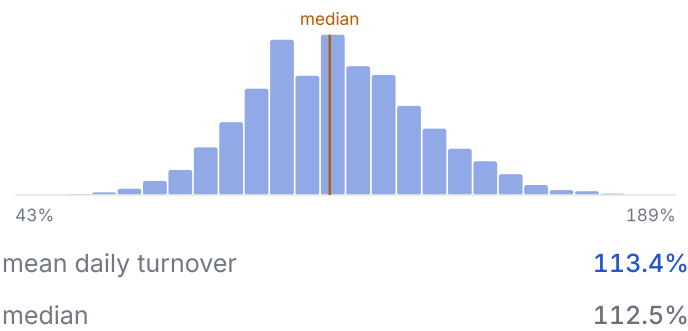
**MSTR.US** Full MASTER



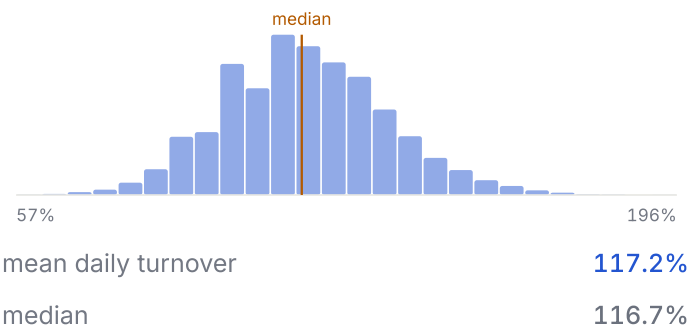
**MSTR.NG** No gating



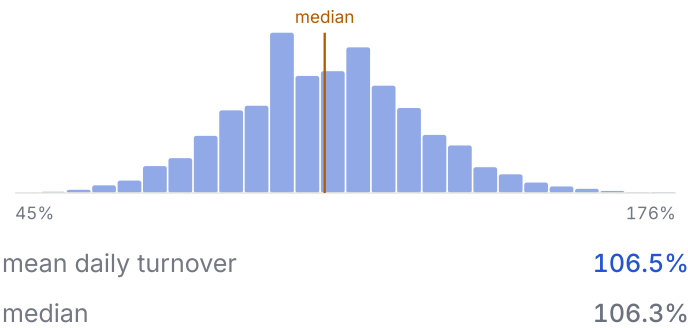
**MSTR.NX** No inter-stock attention



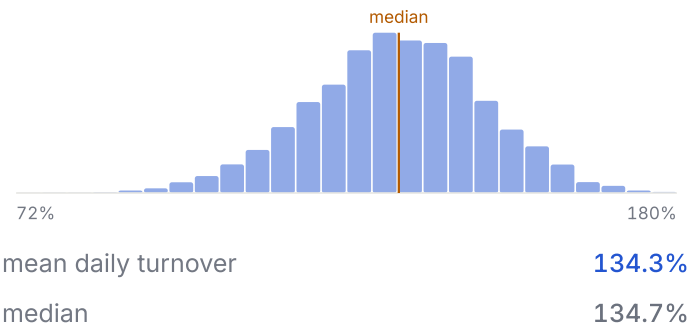
**MSTR.TA** Time-aligned



**MSTR.SH** Market vector shuffled



**LGBM.BL** LightGBM (tuned)



The join — four books under one protocol

5 seeds · test 2019–2025 · net = flat 10 bps

MASTER-US long-short net Sharpe is -0.68. After Barra style-neutralization it is -0.91 — the book carries 77% of its *risk* in style and industry factors while only 15% of its *return* comes from them, so 85% is specific alpha — of which none survives realistic costs, neutralized or not. Cost breakeven occurs at 10.6 bps against a 10 bps assumption.

| BOOK                     |         | GROSS SHARPE  | NET SHARPE    | TURNOVER | BE BPS | VS ZERO         |
|--------------------------|---------|---------------|---------------|----------|--------|-----------------|
| MASTER decile L/S        |         | +0.768 ±0.130 | -0.681 ±0.122 | 116%     | 10.6   | distinguishable |
| optimized, unconstrained | control | +0.189 ±0.103 | -0.624 ±0.120 | 30%      | 2.3    | distinguishable |
| optimized, STYLE-NEUTRAL |         | +0.218 ±0.120 | -0.907 ±0.198 | 31%      | 2.0    | distinguishable |
| same, eigenfactor OFF    | control | +0.218 ±0.120 | -0.907 ±0.198 | 31%      | 2.0    | distinguishable |

A control, not a candidate strategy. Both optimized arms run through the same function with identical lambdas, so the neutral-vs-unconstrained comparison isolates neutralization rather than the switch from a decile rule to an optimizer.

Neutral vs unconstrained net gap **-0.282** — distinguishable. Style-neutralization makes net **worse** (-0.68 → -0.91), because it removes volatility faster than return and the same cost drag falls on a smaller denominator.

Return attribution · gross annualized

\$10.2

|                     |                   |
|---------------------|-------------------|
| total realized      | 15.66%/yr         |
| factor contribution | 2.31%/yr (14.8%)  |
| specific (alpha)    | 13.35%/yr (85.2%) |

NOT AN ALPHA NUMBER

specific gross Sharpe +1.284

This is NOT an alpha number and must not be quoted as one. It is the return left after factor exposure is removed arithmetically, which assumes factor hedging is free. It is not achievable. Constructing the hedge and paying for it is the style-neutral book, which is WORSE than the unhedged book. The gap between the two is the difference between an attribution and a portfolio. Here that hedged book is the style-neutral arm, at **-0.907**.

Risk attribution · predicted variance

\$10.3

76.8%

23.2%

factor share

specific share

The asymmetry is the finding: the book spends **14.61%/yr** of volatility on factor exposure to earn **2.31%/yr** from it — a Sharpe of **0.163** on the factor component. Close to unrewarded risk, and invisible without a factor model.

Factor returns · Phase 5 gate

monthly WLS · post exposure\_lag=1

|                  |                                   |
|------------------|-----------------------------------|
| market           | 14.21%/yr                         |
| momentum         | 0.92%/yr t=1.04 · not significant |
| value            | 0.83%/yr                          |
| mean weighted R² | 0.257                             |

Bias statistic · Phase 6 gate

fraction of portfolios in [0.9, 1.1]

94.6%

|                     |       |
|---------------------|-------|
| in-gate             |       |
| random              | 1.004 |
| factor-mimicking    | 0.939 |
| cap-weighted market | 1.072 |

validates CALIBRATION only — says nothing about factor-return strength

§10.5 Gate interpretation — factor timing

0/8 factors significant · 5 seeds

METHOD NOTE · SUBSTITUTE, NOT IMPLEMENTATION

§10.5 specifies regressing the learned gate activations on the market state vector and on contemporaneous factor returns. No Phase-3 checkpoint was saved and no activations were cached, so the activations are not available. What is measured here instead is the gate's CONSEQUENCE: whether the gated book times factors better than the ungated one. The two models share seeds, data and protocol and differ only in the gate, so the difference is attributable to it. This is a SUBSTITUTE for the specified activation regression, not an implementation of it.

| FACTOR     | MASTER  | UNGATED | GATE Δ  | T (GATED) | SIGNIFICANT |
|------------|---------|---------|---------|-----------|-------------|
| size       | +0.0240 | +0.0064 | +0.0176 | +1.01     | no          |
| value      | +0.0190 | +0.0195 | -0.0005 | +0.80     | no          |
| momentum   | +0.0163 | +0.0206 | -0.0043 | +0.69     | no          |
| volatility | +0.0168 | +0.0132 | +0.0036 | +0.71     | no          |
| liquidity  | +0.0014 | +0.0042 | -0.0028 | +0.06     | no          |
| leverage   | +0.0269 | +0.0022 | +0.0247 | +1.13     | no          |
| growth     | -0.0348 | -0.0222 | -0.0126 | -1.46     | no          |
| quality    | -0.0207 | -0.0176 | -0.0031 | -0.87     | no          |

0/8  
significant

+0.0247  
largest Δ · leverage

0.0211  
seed dispersion

0/8 factors show significant timing. Largest gate delta +0.0247 (leverage) against seed dispersion 0.0211 — inside the noise. The gate does no detectable regime-conditional factor timing. Together with Phases 3–4 (the gate does not predict returns distinguishably better than no gating), this completes the null: a mechanism that neither improves prediction nor times factors.

Eigenfactor prediction check

NOT confirmed

+0.0002  
net Sharpe gap · on vs off

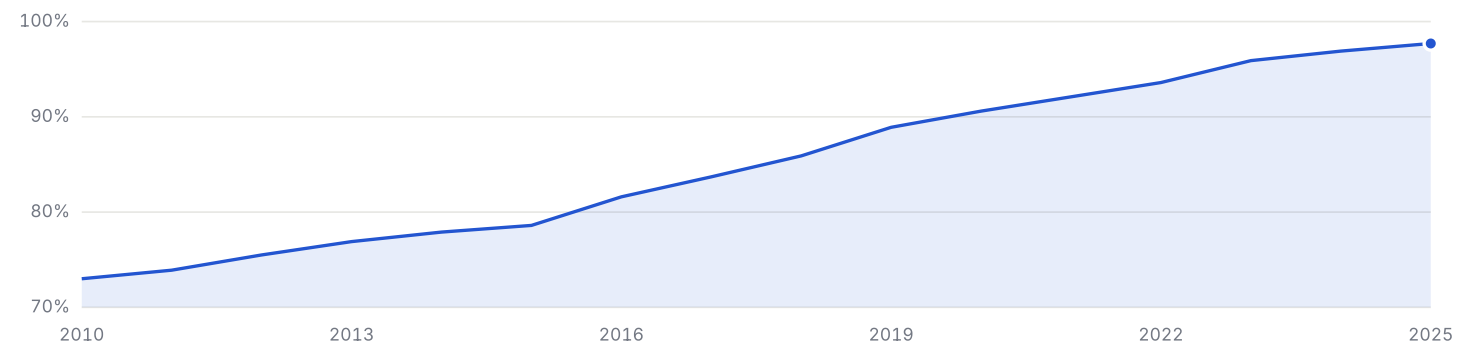
NOT DISTINGUISHABLE  
verdict

not distinguishable  
failed prediction

Phase 6 found the eigenfactor adjustment slightly hurt the bias statistic and argued it should help once a real optimizer ran against the covariance, because the adjustment corrects minimum-variance directions while Phase 6's test portfolios were random and factor-mimicking, neither optimized. Phase 7 ran that optimizer. The prediction is NOT confirmed, and is recorded as a failed prediction rather than explained away.

Survivorship — what the free-data path costs

613 / 795 historical constituents retrievable · 77.1%



Retrieval climbs monotonically from **73.0%** in 2010 to **97.7%** in 2025. The early sample is the most contaminated — and the early sample is the training set. Direction of bias: **OPTIMISTIC**. Measured, not corrected; the free path cannot fix it, so it is stated instead.

Coverage by year

| YEAR | CONSTITUENTS | RETRIEVED | RATE  |
|------|--------------|-----------|-------|
| 2010 | 512          | 374       | 73.0% |
| 2011 | 517          | 382       | 73.9% |
| 2012 | 519          | 392       | 75.5% |
| 2013 | 516          | 397       | 76.9% |
| 2014 | 516          | 402       | 77.9% |
| 2015 | 529          | 416       | 78.6% |
| 2016 | 534          | 436       | 81.6% |
| 2017 | 534          | 447       | 83.7% |
| 2018 | 524          | 450       | 85.9% |
| 2019 | 521          | 463       | 88.9% |
| 2020 | 521          | 472       | 90.6% |
| 2021 | 520          | 479       | 92.1% |

| YEAR | CONSTITUENTS | RETRIEVED | RATE  |
|------|--------------|-----------|-------|
| 2022 | 519          | 486       | 93.6% |
| 2023 | 515          | 494       | 95.9% |
| 2024 | 516          | 500       | 96.9% |
| 2025 | 518          | 506       | 97.7% |

Protocol

|                |           |
|----------------|-----------|
| Train          | <=2016    |
| Valid          | 2017-2018 |
| Test           | 2019-2025 |
| Embargo        | 21 d      |
| Seeds required | 5         |
| Cost baseline  | 10 bps    |

Chronological splits, no shuffling and no k-fold. Fundamentals join on the SEC `filed` date, never `period_end` . Normalization statistics are fit on train only and borrowed by valid and test.

Sources

|                |                                                    |
|----------------|----------------------------------------------------|
| Metrics bundle | data/processed/phase4/metrics_bundle.npz           |
| Scores         | data/processed/phase2/<variant>_seed<n>_scores.npy |
| Phase results  | reports/status/phase*.json                         |
| Survivorship   | reports/survivorship.md                            |

Deferred — not run, not estimated see methodology

§8.1 ROW 6 · DEFERRED — NOT RUN · 70-95 H PROJECTED  
Lookback sweep — L = 40 / 60 / 120, both arms  
Deferred rather than run at reduced power. No results exist.

§8.1 ROW 7 · DEFERRED — NOT RUN · ~12.2 H PROJECTED  
Head sweep — (4,2) / (8,8) / (16,4), both arms  
Launched and stopped after 1 of 30 runs; that output was deleted. No results exist.

Neither affects the gate-null conclusion, which was established at the default configuration with 10 confirmatory seeds at the spec's full budget, a 25-run  $\beta$  sweep, and a market-shuffle control. A sweep could only show the gate helps at some *other* hyperparameter setting — a new claim carrying its own evidential burden.

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Every figure is computed from committed artifacts — per-seed score arrays, `reports/status/phase*.json`, `reports/phase7_tables.json`, and `reports/survivorship.md`. Nothing on this page is illustrative.